

AL SYSTEM 

Challenge & Strategic Mission



The Problem

Traditional systems rely on **fixed timing**, which fails in



The Mission

Develop an **autonomous intelligent agent** that

Challenge & Strategic Mission



The Problem

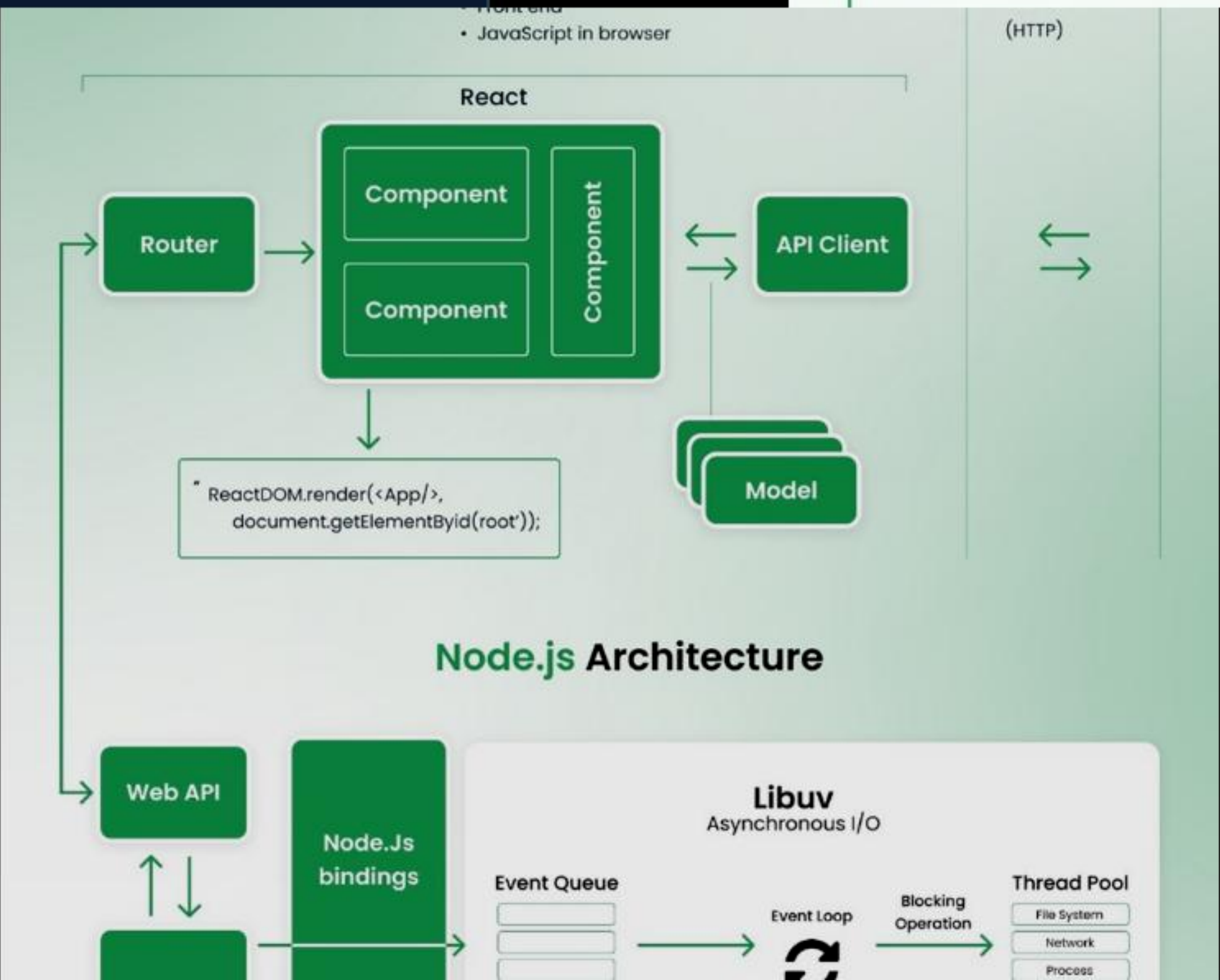
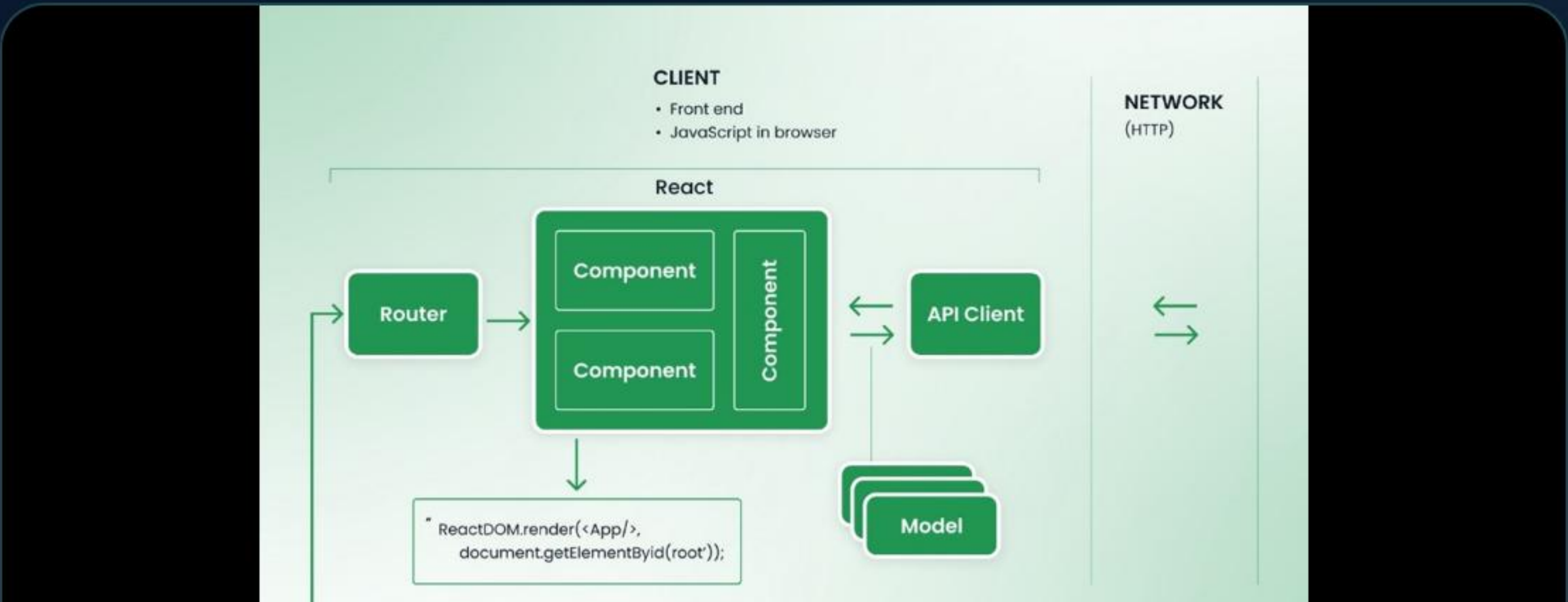


The Mission

System Architecture & Tech Stack

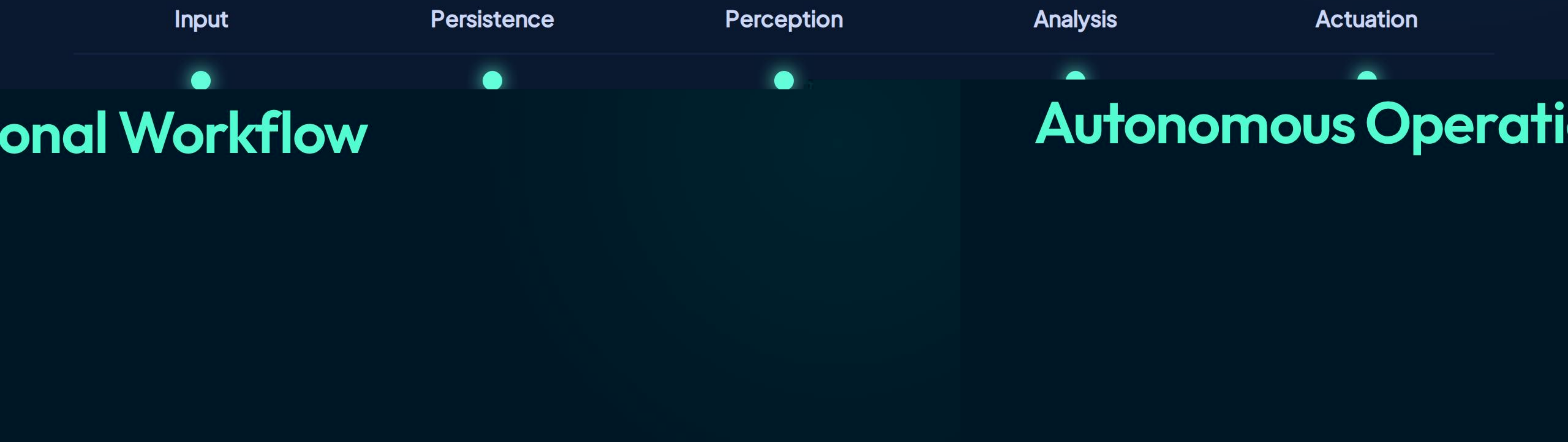
Full-Stack Integrated Core

-  **Frontend:** React for intuitive event management
-  **Backend:** Node.js + Express API bridge
-  **Storage:** MongoDB NoSQL for persistence
-  **Intelligence:** Python-based Decision Engine



-  **Frontend:** React
 -  **Backend:** Node
 -  **Storage:** Mongo
 -  **Intelligence:** Py
- Python acts as the central 'brain', processing temporal data and state tracking.*

Autonomous Operational Workflow



Agent Logic: Utility-Based Reasoning

Threshold Logic Formula:

$$T_{\text{alert}} = T_{\text{event}} - \rho (\text{Priority})$$

PRIORITY LEVEL	LEAD TIME (P)	LOGIC STRATEGY	SYSTEM RESPONSE
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Intelligent Features & Testing

Adaptive Timing

Calculates proximity based on urgency, not fixed intervals.

State Tracking

Threshold logic ensures alerts fire only once —no duplicate spam.

Self-Cleaning

Automatically purges past events to maintain DB performance.

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Intelligent Features & le

king

Researcher: 100% accurate and 2 seconds to run on 1000 items.

Self-Cleaning

200 items deleted every 2 weeks to keep DB size small and query performance high.

Adaptive Timing

Alerts are sent only if the user has not seen the alert within 24 hours.

State Tracking

Alerts are only sent once per user.



Test Validation:

Engineering Challenges & Roadmap

Challenges Resolved

- ✗ **CORS Issues:** Handled cross-origin resource sharing for JS-Python calls.
- ✗ **Time Delta:** Resolved 'T' vs 'Space' ISO format mismatches.
- ✗ **Async Loop:** Debugged repeated notification bugs in

Future Enhancements

- **WebSockets:** Real-time socket-based notification delivery.
- **Conflict Detect:** Auto-detection of overlapping schedule priorities.

me socket-based notification

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✗ **CORS Issues:** Handled cross-origin resource sharing for JS-Python calls.

✗ **Time Delta:** Resolved 'T' vs 'Space' ISO format mismatches.

✗ **Async Loop:** Debugged repeated notification bugs in the main agent loop

➤ **WebSockets:** Real-time socket-based notification delivery.

➤ **Conflict Detect:** Auto-detection of overlapping schedule priorities.

➤ **ML Logic:** Learning user habits to auto-set priority levels

Any Questions?

Smart Autonomous Event Reminder Agent

VIVA KEY POINTS:

- Model: Utility-Based Intelligent Agent
- Cycle: Sense (DB) → Analyze (Python) → Decide (Threshold) → Act (Notify)
- Core Innovation: Threshold-crossing state tracking

Any Questions?

Smart Autonomous Event Reminder Agent

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Image Sources



https://blogger.googleusercontent.com/img/b/R29vZ2xl/AVvXsEim_e2EGxs04ylBsoHRC9WXspi6-5X8DUOqIVQxn8uWPjqQKVmi4rvrOnXrVJp6WP1yKVnPOqHNN2dmdlOR2Ilm5nX3di5Fr62iT7YZLEE_8Tv7p7VZU-eyzjrTeuw-50robYWt_NVsbQTWQgL2x4Mc8ls86l37yHbLSvZ5u8dafnDl16JjjQ6xOLU5Soq/s1100/Imagen2.jpg

Source: www.itarch.info